

Science and Archaeology

What, then, is the relationship between archaeology and the natural sciences, such as biology, chemistry, physics, and geology? From its earliest days, archaeology has been closely connected with these scientific disciplines. For the most part, archaeologists have borrowed techniques from these scientific disciplines, but in some cases, unique methods were developed first for archaeology and later found broader applications in science.

Geology

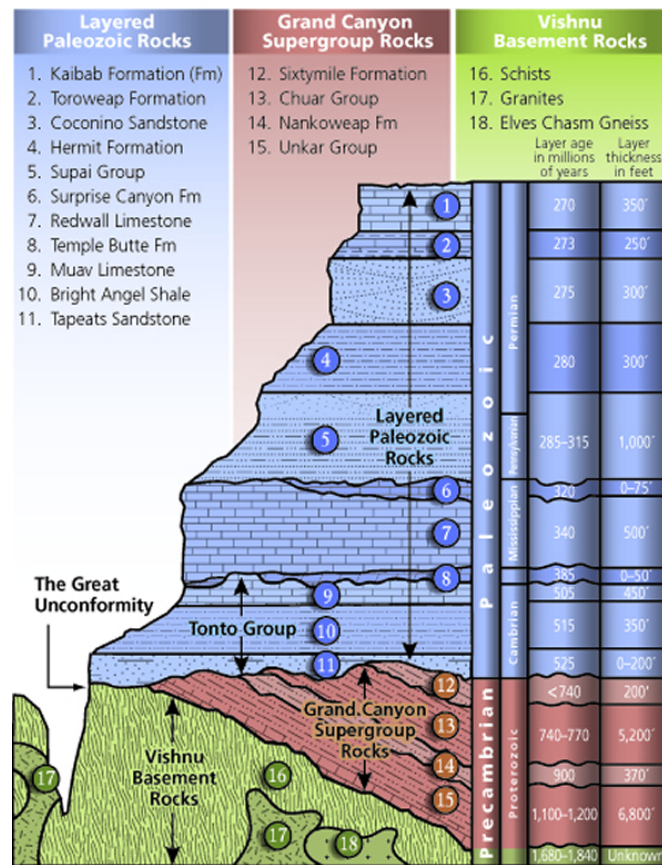
Archaeologists rely on some of the basic principles behind geology—especially the principle of **uniformitarianism**, which holds that processes that we can observe in the present (such as volcanism, sedimentation, and erosion) operated the same in the past, so that the geological formations we see today are simply the result of the same processes operating very slowly over very long periods of time. Another key principle from geology that shaped archaeology is that of superposition, which stipulates that younger layers of rock and soil form over older layers. If we look at a series of rock layers, such as those seen in a profile like that of the Grand Canyon, the oldest layers should lie at the bottom and the youngest on the top. Archaeologists use this principle to determine which layers of an archaeological site are older and which are younger, which helps us to date specific layers and their associated artifacts and ecofacts.



The Grand Canyon, Arizona.

Source: Dr. Marston

Grand Canyon's Three Sets of Rocks



Stratigraphic diagram of the Grand Canyon.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Stratigraphy_of_the_Grand_Canyon.png)



Archaeological stratigraphy preserved in the Acropolis metro (subway) station, Athens, Greece.

Source: [Wikimedia Commons](#)

(https://commons.wikimedia.org/wiki/File:Acropolis_Metro-Station.JPG)

Uniformitarianism is the principle that geological processes visible in the present day are those that operated in the past. For example, volcanos such as we find today created mountain ranges, which were then worn down by millennia of erosion from wind, rain, and glacial activity. These are the same geological processes that exist in the present.

Superposition is the principle that items and layers (strata) found underneath other layers predate those above them, absent postdepositional disturbance. A stack of mail that accumulates on a desk will have the oldest on the bottom and most recent on the top; a floor may have multiple layers of linoleum tiles atop one another, and those on the bottom will be the earliest. It is the same with geological layers and archaeological layers, unless someone or something has dug into the deposits and removed material, which has later been filled in with more recent material.

Chemistry and Physics

As techniques in chemistry and physics have developed since the 19th century, archaeologists have adopted useful tools from these fields to develop methods that aid in answering archaeological questions. We will see later in the course several examples of these techniques, such as chemical-residue analysis, which allows archaeologists to find out what substances were stored in an ancient pottery jar or cooked in a casserole dish. One ubiquitous tool in archaeology is **radiocarbon dating**; Willard Libby was awarded the Nobel Prize in Chemistry in 1960 for the development of this method based on radioactivity research conducted as part of the Manhattan Project in the 1940s. This is still the only Nobel Prize ever awarded for an archaeological discovery.

Radiocarbon dating (also termed "carbon-14 dating") is a method for determining the period of time between the death of an organism and the present. This tool allows us to date various archaeological finds, primarily ecofacts, as well as human skeletons. The basic principle of the technique is that a small percentage of carbon atoms on Earth (about 1 in 10 million) are a radioactive **isotope**, carbon-14, that is heavier than the more common, stable isotopes of carbon (carbon-12 and carbon-13; the number refers to the atomic weight of the isotope). Being radioactive, this isotope is unstable and breaks down into nitrogen-14 on a probabilistic basis. Although we cannot predict when any one atom will decay, we know that, on average, half of all carbon-14 atoms in an object will break down after one half-life—here, 5,730 years. After two half-lives (11,460 years), only one-quarter of the original radiocarbon will remain; after four (22,920 years), only one-sixteenth of the original radiocarbon will remain. After 10 half-lives (57,300 years), less than 1/1,000 of the original radiocarbon will still be present, which approaches the typical limit for detection. This means that substances older than about 60,000 years cannot be dated using radiocarbon dating—there is just not enough radioactive carbon left to detect. The other major limitation of the technique is that it only can be used on materials that originally contained organic carbon—that is, organic materials. These include any part of a once-living being (e.g., bone, tooth, hair, plant fiber, seeds, wood, clam shells) but not stone, clay, metal, or other inorganic substances. Thus, while we can date fragments of charcoal from a fire pit or bones from a burial, we cannot directly date most artifacts (e.g., pottery, metal

tools) or buildings, as only stone or earth components of their structures usually survive.

Isotopes are different forms of atoms of an element, determined by their molecular weights (their total numbers of protons and neutrons). All atoms belong to one element and one isotope. Radioactive isotopes are unstable and will change into another isotope, usually of another element. Stable isotopes remain indefinitely unless acted upon by an outside force.

Test Yourself

If a seed from an archaeological site contains one-eighth the amount of carbon-14 present in a living seed, how old is it?

☐ A - About 5,700 years old

☐ B - About 11,400 years old

☒ C - **Correct:** About 17,200 years old Correct answer

☐ D - About 22,900 years old

About 17,200 years old. If a substance has lost half of its carbon-14, it is one half-life old. If it has lost $\frac{3}{4}$, it is two half-lives old. If it has lost $\frac{7}{8}$, it is three half-lives old. Three half-lives is $3 \times 5,730 = 17,190$. Answer c is the closest answer.

Biology

Techniques borrowed from biology are critical to the archaeological study of food. We rely on knowledge from centuries of botanical research to know what plant a specific seed found archaeologically came from, and on knowledge of comparative osteology—the study of bones—to distinguish the bones of one fish from another when the bones are found in an ancient trash deposit. Modern techniques in molecular biology allow archaeologists to study DNA and proteins from a variety of substances, from ancient bones of people and animals to plant fibers used to make clothing found buried in tombs. These methods allow archaeologists to determine not only what plant or animal was used by people in the past but also some of the characteristics of that plant that do not preserve archaeologically, such as whether an ear of corn was sweet or starchy.